

Question Bank

Unit 1: Water Technology

Section A: Short Answer Questions (2 marks each)

1. Define hardness of water. Distinguish between temporary and permanent hardness.
2. What is the principle of the EDTA method for the determination of hardness?
3. Explain why boiler feed water needs treatment.
4. What is Calgon conditioning? How does it help in water treatment?
5. Differentiate between internal and external water treatment methods.
6. What is zeolite? How does it help in softening of water?
7. Write the chemical reaction involved in lime-soda process.
8. What are boiler scales? Mention any two ill effects of scale formation.
9. What is phosphate conditioning?
10. Define ion exchange process.

Section B: Long Answer / Descriptive Questions (5 marks each)

11. Explain the EDTA method for the determination of hardness of water with chemical equations.
12. Discuss internal treatment of boiler feed water with reference to phosphate and colloidal conditioning.
13. Explain the principle, construction, and working of the zeolite process for softening of water.
14. Compare lime-soda process and ion exchange process for softening of water.
15. Explain the formation of boiler scales. What are the ill effects of scale formation in boilers?
16. Discuss methods to prevent scale and sludge formation in boilers.
17. What are the advantages and disadvantages of internal conditioning methods for boiler water treatment?
18. Explain in detail the external water treatment methods.
19. Write a short note on colloidal conditioning and its role in water treatment.
20. Discuss the importance of boiler feed water treatment in industries.